

Study and Implementation of Self-Balancing Electric Motorcycle

The Dang Phan, Chau Binh Pham, and Cong Bang Pham

Ho Chi Minh City University of Technology, Ho Chi Minh City, Viet Nam

Abstract. Self-balancing electric motorcycles have been recently developed by high-tech corporations to reaffirm their vision of “enjoying the freedom of mobility”. A stabilization system based on Inertia Wheel Pendulum (IWP) have been proven to be a potential approach to this new concept. This paper describes the mathematical models of the motor and the vehicle. Sequentially, the effects of some key parameters onto the model are studied to properly design the reaction wheel. Moreover, IMU sensor performance with Kalman filter has been successfully implemented. PID and LQR control algorithms have been explored through simulation. Consequently, the result has indicated that the LQR controller provides more stable responses. In addition, LQR algorithm has been validated on the physical prototype in various scenarios.

Keywords: Inertia Wheel Pendulum · Self-balancing vehicle · LQR controller.

1 Introduction

Along with the rapid advancement of science and technology, the demand for convenient means of transportation is showing the emergence of many ideas for small, fast-moving devices to replace walking. Since 2001, self-balancing two-wheeled vehicles - considered a revolutionary type of personal transportation - have begun to be widely introduced. With that idea, Segway developed the i2 SE electric scooter that weighs 48 kg, achieves a maximum speed of 20 km/h, and is able travel a distance of approximately 34 km. Self-balancing two-wheeled vehicles are operated on the principle of the inverted pendulum. A person standing on the vehicle is represented by a rod of length l and mass m . When the rod tilts forward at an angle α , the control system sends a control signal to push the vehicle forward, causing the angle α to approach 0° and therefore the vehicle is balanced vertically.

The idea of self-balance was later developed into two-wheeled motorbikes. Vehicle corporations have continuously developed and introduced numerous models of self-balancing electric motorcycles, exemplified by models from Lit Motors with the product launched in 2016, and from Honda that was first introduced in 2017 at the Consumer Electronics Show (CES) in Las Vegas. Similar to the balancing system of an inverted pendulum, self-balancing electric motorcycles

utilize a sensor to determine the tilt angle and an actuating mechanism to generate the required balance. Over the past few decades, researchers have proposed various balancing mechanisms grouped into three major methods with heading steering, mass moving, and flywheel.

In balancing by heading steering, a steering motor is mounted on the bicycle's steering mechanism to change the heading angle in order to generate a centrifugal force to balance the bicycle. Yi has developed a mathematical model for the autonomous motorcycle [1]. Using this balancing method normally cannot be balanced at zero speed. In balancing by mass moving, a pendulum is mounted on the bicycle's frame and is actuated in order to keep the center of gravity of the whole system on the line between both wheel-ground contact points so that the bicycle can remain upright. Seekhao has studied the integration of steering and a non-inverted pendulum in balancing the bicycle at both zero and low speeds [2]. However, using mass moving requires a wide pendulum-swinging range. Using flywheel, in which one or more flywheels are mounted on the bicycle's rear frame in various ways to generate a torque against the bicycle's falling, has been found a potential approach to overcome disadvantages from the previous mechanisms. This has attracted interests from many researchers under two common terms: gyroscope and inertial wheel.

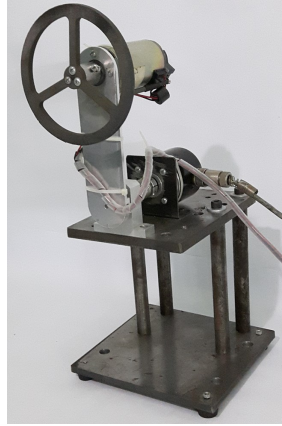


Fig. 1. Physical model of the IWP

For the gyroscope, using CMG (Control Moment Gyroscope) has been explored and used as stability control of a bike with various degrees of success [3]. An LQR (Linear Quadratic Regulator) controller has been demonstrated to provide a good performance such as less vibration, faster response, and greater robustness [4]. For the inertial wheel, a common method of attitude control for spacecraft, the conservation of angular momentum of the wheel has also been explored to stabilize a bike [5]. A reinforcement learning technique has been pro-

posed in this mechanism in term of energy efficient [6]. A conventional controller of PID is also applied to keep a bike vertical stably [7]. A prototype of IWP, as shown in Fig. 1, has also built to demonstrate its stable condition using a PID controller [8].

So far, the self-balancing electric motorcycle is still attracting the interest of many people and opening various research directions as well as practical applications. Therefore, with the purpose to apply the IWP above on self-balancing electric motorcycles, a prototype of self-balancing motorcycle will be developed in this paper. Moreover, various control algorithms such as PID and LQR will be explored through simulations and experimental validation. The organization of this paper is as follows: Section 2 highlights mathematical models of the motor and the vehicle; and mainly presents on effect of key parameters on system response. This is followed by steps to develop a prototype of a self-balancing motorcycle in Section 3. A comparison between PID and LQR algorithms is discussed through Matlab simulation in Section 4. Subsequently, Section 5 covers a comprehensive LQR controller that is validated on the prototype under various scenarios. The paper is summarized in Section 6.

2 System Modeling

In general, a self-balancing motorcycle is a system whose working principle can be described as an IWP as shown in Fig. 2. There are two main components that need proper modeling which are the balancing motor (1) and the vehicle frame (2).

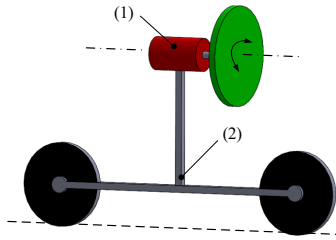


Fig. 2. 3D model of IWP

2.1 Modeling DC motor

Assume that the DC motor is utilized to self-balance the system. Therefore, the relationship between input voltage and output torque of the motor will be illustrated below.

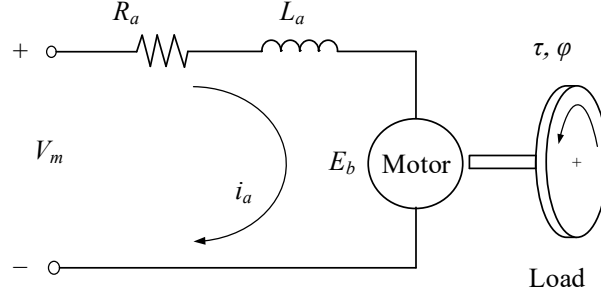


Fig. 3. DC motor model

The equivalent electric circuit of the DC motor and the free-body diagram of the load are shown in Fig. 3 with R_a , L_a represent the armature winding resistance and inductance. Applying Kirchhoff's second law leads to:

$$V_m = i_a R_a + L_a \frac{di_a}{dt} + E_b \quad (1)$$

The back electromotive force (emf) is related to the angular velocity θ by the back-emf constant K_b as:

$$E_b = K_b \dot{\phi} \quad (2)$$

Assuming the effect of inductance is very small ($L_a \ll R_a$), Eqn. (1) becomes:

$$i_a = \frac{V_m - K_b \dot{\phi}}{R_a} \quad (3)$$

The motor torque τ is related to armature current i_a by a motor torque constant K_t as:

$$\tau = K_t i_a = K_t \frac{V_m - K_b \dot{\phi}}{R_a} \quad (4)$$

2.2 Modeling inverted wheel pendulum

The vehicle frame consists of a pendulum and an inertia wheel as shown in Fig. 4. In detail, the pendulum can rotate freely about the Oz axis, which is perpendicular to the Oxy plane. Besides, the inertia wheel is driven by the DC motor to rotate about the axis $O'z$ which is parallel to the Oz axis. When the pendulum falls off, the speed of a reaction wheel is increased or decreased to generate a reactionary torque to self-balance the system.

A mathematical model of the vehicle frame is established using Lagrange method. Energy of system consists of kinetic and potential energy as follows:

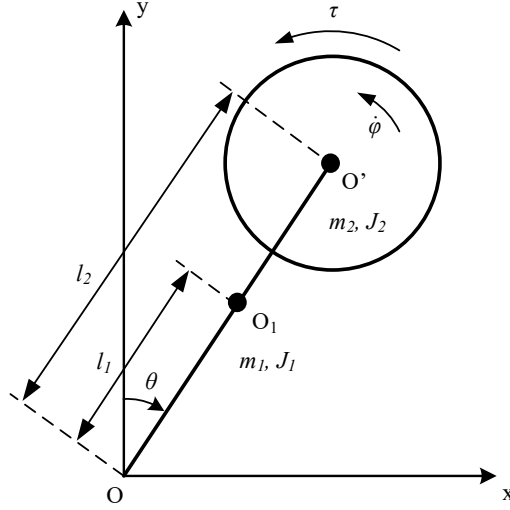


Fig. 4. IWP model

- Kinetic energy of the system, including translational parts and rotational parts:

$$T = \frac{1}{2}(m_1 l_1^2 + m_2 l_2^2 + J_1 + J_2)\dot{\theta}^2 + J_2 \dot{\theta} \dot{\phi} + \frac{1}{2} J_2 \dot{\phi}^2 \quad (5)$$

- Potential energy of the system:

$$V = (m_1 l_1 + m_2 l_2)g \cos \theta \quad (6)$$

The Euler-Lagrange equation with a dissipated energy is expressed in Eqn. (7).

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} + \frac{\partial R}{\partial \dot{q}_i} = 0 \quad (7)$$

Where

- $L = T - V$ is the Lagrangian of the system

$$L = \frac{1}{2}(m_1 l_1^2 + m_2 l_2^2 + J_1 + J_2)\dot{\theta}^2 + J_2 \dot{\theta} \dot{\phi} + \frac{1}{2} J_2 \dot{\phi}^2 - (m_1 l_1 + m_2 l_2)g \cos \theta \quad (8)$$

- R is the Rayleigh dissipation function

$$R = \frac{1}{2} c_1 \dot{\theta}^2 + \frac{1}{2} c_2 \dot{\phi}^2 \quad (9)$$

- q_i are the generalized coordinates

Applying Eqn. (7) for the angle of vehicle frame θ and the angle of inertial wheel φ leads to Eqns. (10) and (11), respectively.

$$(m_1 l_1^2 + m_2 l_2^2 + J_1 + J_2) \ddot{\theta} + J_2 \ddot{\varphi} + c_1 \dot{\theta} - (m_1 l_1 + m_2 l_2) g \sin \theta = 0 \quad (10)$$

$$J_2(\ddot{\theta} + \ddot{\varphi}) + c_2 \dot{\varphi} = \tau \quad (11)$$

Substituting Eqn. (4) into Eqn. (11) results in:

$$J_2(\ddot{\theta} + \ddot{\varphi}) + c_2 \dot{\varphi} = K_t \frac{V_m - K_b \dot{\varphi}}{R_a} \quad (12)$$

From Eqns. (10) and (12), the mathematical model of the self-balancing vehicle is given in Eqn. (13) with the notation of parameters are summarized in Table 1.

$$\begin{bmatrix} m_1 l_1^2 + m_2 l_2^2 + J_1 + J_2 & J_2 \\ J_2 & J_2 \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\varphi} \end{bmatrix} + \begin{bmatrix} c_1 & 0 \\ 0 & \frac{K_t K_b}{R_a} + c_2 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\varphi} \end{bmatrix} + \begin{bmatrix} -(m_1 l_1 + m_2 l_2) g \sin \theta \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{K_t}{R_a} \end{bmatrix} V_m \quad (13)$$

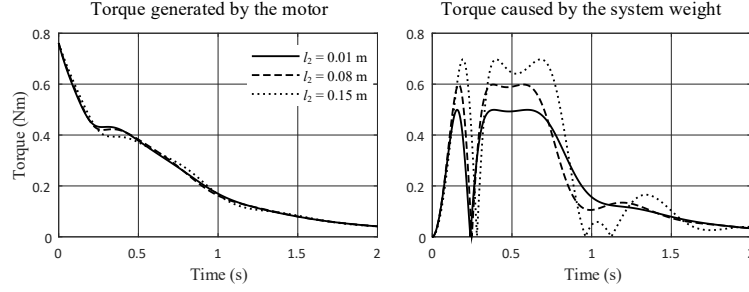
Table 1. Parameter notations of the IWP

Symbol	Unit	Definition
θ	rad	Angle of pendulum
φ	rad	Angle of inertia wheel
J_1	kg.m ²	Pendulum inertia moment including stator
J_2	kg.m ²	Wheel inertia moment including rotor
c_1	Nm.s/rad	Friction factor of pendulum
c_2	Nm.s/rad	Friction factor of wheel
m_1	kg	Mass of pendulum and stator
m_2	kg	Mass of wheel and rotor
l_1	m	Length from origin to pendulum center gravity
l_2	m	Length from origin to wheel center gravity
K_b	V/(rad/s)	Back electromotive force constant
K_t	Nm/A	Motor torque constant

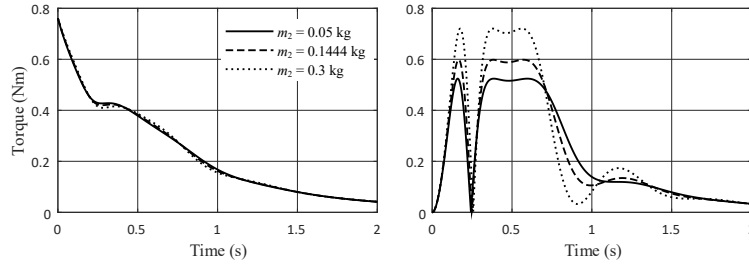
2.3 The effects of certain mechanical parameters on the system performance

A preliminary design is assembled in SolidWorks to obtain nominal values of the parameters. In this system, three key parameters l_2 , m_2 and J_2 have significant

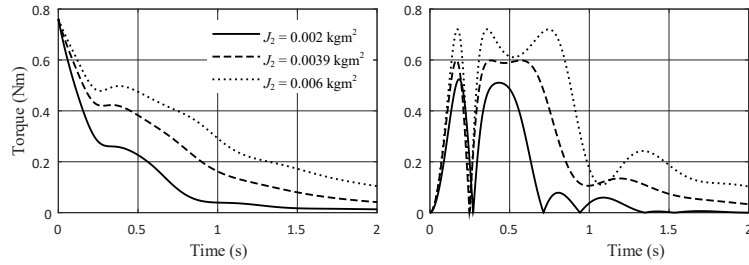
effects on the balancing ability in the system. Two quantities of torque generated by motor and torque caused by system weight are observed under various values of these three parameters when applying step voltage inputs.



(a) Response under different l_2



(b) Response under different m_2



(c) Response under different J_2

Fig. 5. Torque responses

With the vehicle hold upright initially, torques from the motor cause the system to fall downwards with respect to time. As shown in Fig. 5, figures in the left column demonstrates the torque generated by motor acting on the system,

while figures in the right column describes the absolute required torque to keep its balance under the gravity. It is clearly observed that:

- Changing l_2 as well as m_2 does not affect the output torque of the motor.
- Increasing either l_2 or m_2 results in larger amount of torque caused by the weight of the motorcycle.
- The larger J_2 is, the greater motor torque is created.
- When J_2 is increased, with larger m_2 , it is clear that the required torque in the system is also higher.

3 Prototype Development

3.1 Mechanical design

There are four main components for mechanical structure: reaction wheel, balancing motor, driving component and steering mechanism.

- For the reaction wheel, according to the analysis above, it needs to have higher inertia moment and smaller mass.
- The balancing motor needs to be able to generate the sufficient amount of torque and maintain it for a long period to keep the motorcycle in balance.
- The driving torque is calculated in order for the vehicle to move forward with the speed of 0.1 m/s.
- The steering mechanism is designed so that the steering angle is up to 40° .

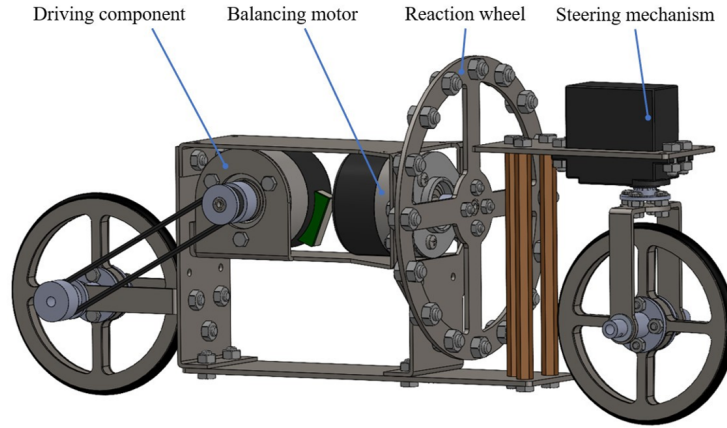


Fig. 6. 3D model of self-balancing motorcycle

Figure 6 demonstrates a 3D model of self-balancing motorcycle. The wheel has multiple pairs of bolt and nut installed on the outer ring with radius of 55 mm. Nidec motor 24H404H160 is selected for both balancing and driving component. A servo motor MG996R is chosen in the steering mechanism.

3.2 Electrical design

The electrical system is demonstrated in Fig. 7 in which, Atmega2560 is used as a controller, inertia measurement unit (IMU) MPU6050 is utilized to sense the tilt angle. Four battery cells power source of 14.8 V is regulated into 7 V for the micro-controller and steering motor, and into 12 V for balancing and driving motors.

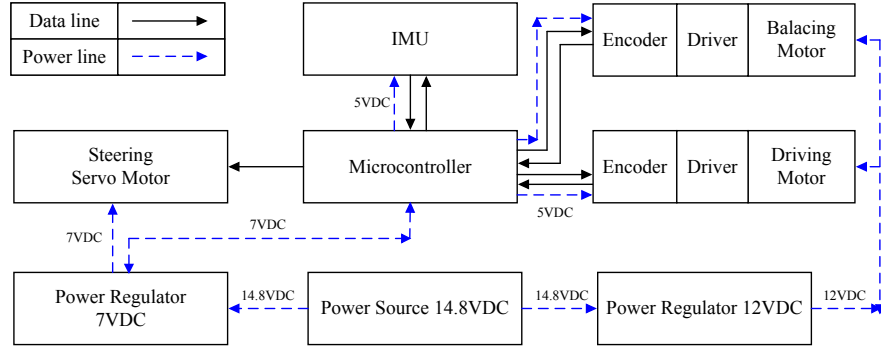


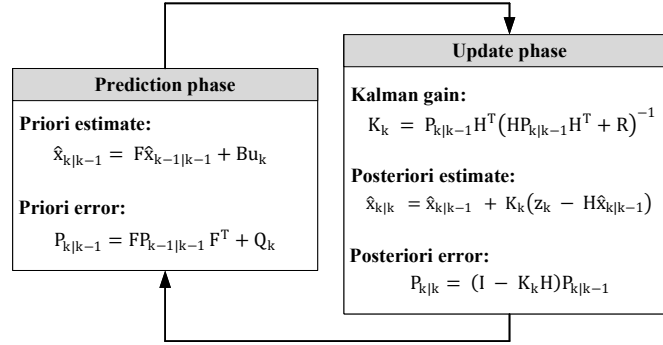
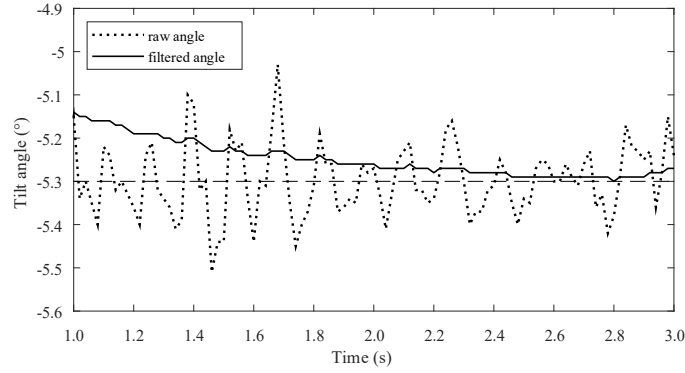
Fig. 7. Electrical system overview

3.3 Tilt angle estimation using Kalman filter

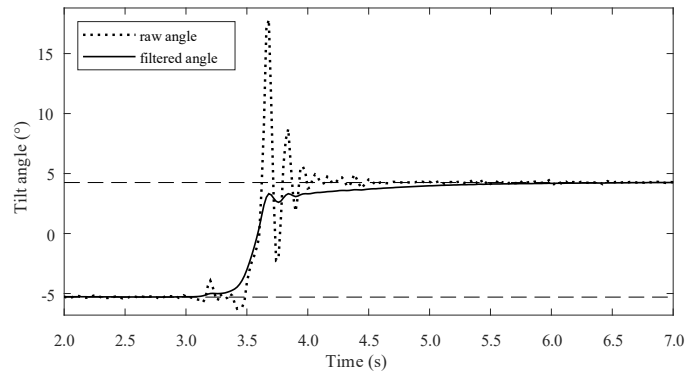
The performance of the self-balancing motorcycle model is significantly influenced by its sensing system. For the MPU6050, its raw angles are strongly affected by noise. Hence, a Kalman Filter is used to obtain usable tilting angle. Basically, Kalman filter algorithm works recursively by a two-phase process as shown in Fig. 8. For the prediction phase, the Kalman filter produces estimates of the current state variables, along with their uncertainties. Once the outcome of the next measurement (necessarily includes with some errors and random noise) is observed, these estimates are updated using a weighted average, with more weight being given to estimates with greater certainty.

Fig. 9 presents the tilt angles measured in two scenarios: when the system is stationary and when the system is tilted from side to side.

- As seen in Fig. 9(a) in which the system is kept at a static approximate angle of -5.3° , the filtered angle is gradually converged to this static angle from an initial Kalman value set at 0° whereas the raw angles always jump up and down around this value.
- As seen in Fig. 9(b) in which the system is approximately swung from -5° to $+5^\circ$, the filtered angle also responds smoother than the raw angle. Therefore, the filtered angle is better for a control system.

**Fig. 8.** Kalman Filter algorithm loop

(a) When the system is stationary



(b) When the system is swung side to side

Fig. 9. Tilt angle measurement

4 Simulation Results

After obtaining the necessary parameters from the mechanical models previously designed, the system was tested in a simulation environment to evaluate its balancing performance when using controllers, such as the PID controller and LQR controller.

4.1 Control flowchart

According to (13), the model for the prototype is non-linear due to the appearance of the function $\sin \theta$. Because when the prototype is self-balancing, its tilt angle only oscillates within the range of $\pm 3^\circ$. Therefore, the mathematics model can be simplified with $\sin \theta \approx \theta$. As a result, the linear model can be achieved as follows:

$$\begin{bmatrix} m_1 l_1^2 + m_2 l_2^2 + J_1 + J_2 & J_2 \\ J_2 & J_2 \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\varphi} \end{bmatrix} + \begin{bmatrix} c_1 & 0 \\ 0 & \frac{K_t K_b}{K_a} + c_2 \end{bmatrix} \begin{bmatrix} \dot{\theta} \\ \dot{\varphi} \end{bmatrix} + \begin{bmatrix} -(m_1 l_1 + m_2 l_2) g \theta \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{K_t}{R_a} \end{bmatrix} V_m \quad (14)$$

Implementing a PID controller on the self-balancing motorcycle, the actuating tilt angle θ_a of the prototype is the feedback signal to determine the error with the set point desired θ_d . Fig. 10 demonstrates the PID controller block diagram of the system. The PID coefficients are preliminary tuned with $K_p = 100$, $K_i = 100$ and $K_d = 2$.

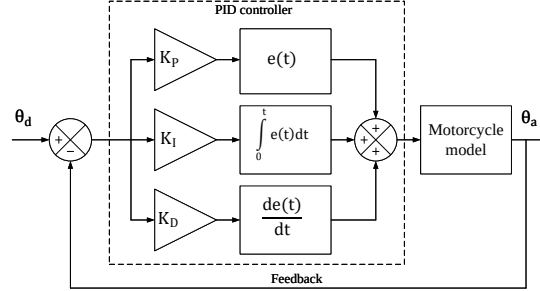


Fig. 10. PID controller block diagram

Different than the PID controller, LQR is a linear control system which considers the effects of more input parameters, including model tilt angle θ , tilt velocity $\dot{\theta}$, reaction wheel rotational angle φ and velocity $\dot{\varphi}$ of the system as shown in Fig. 11. The control law derived from the LQR design is given by:

$$u = -Kx \quad (15)$$

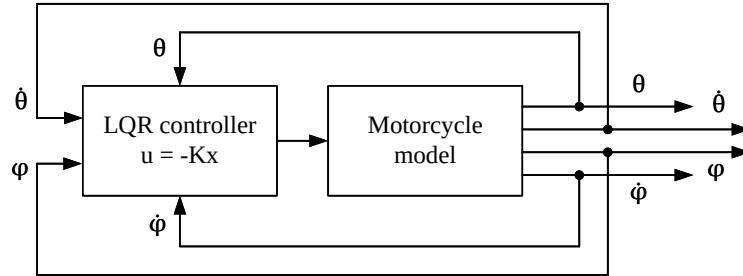


Fig. 11. LQR controller block diagram

The feedback gain matrix K is derived as (16).

$$K = R^{-1} B^T S \quad (16)$$

Where S is the positive definite (or positive semi-definite) solution of the algebraic Riccati equation, solved from (17).

$$0 = SA + A^T S - SBR^{-1}B^T S + Q \quad (17)$$

The control weighting matrix Q can be described below:

$$Q = \begin{bmatrix} Q_{11} & 0 & 0 & 0 \\ 0 & Q_{22} & 0 & 0 \\ 0 & 0 & Q_{33} & 0 \\ 0 & 0 & 0 & Q_{44} \end{bmatrix} \quad (18)$$

Firstly, the state weighting parameters Q_{11} , Q_{22} , Q_{33} , and Q_{44} are adjusted in such a way that the wanted state will have higher priority compared with other states. In addition, to simplify the tuning process, the weight for energy consumption R will not be considered, therefore, it is set to 1. After that, the feedback gain matrix K is calculated with respect to each control weighting matrix R and each state weighting matrix Q , and then used to evaluate the system performance. As a result, the LQR coefficients are also preliminary tuned with $Q_{11} = 4000$, $Q_{22} = 150$, $Q_{33} = 1$, $Q_{44} = 5$, and $R = 1$.

4.2 Performance responses

To test the performance of the system, the responses will be observed and analyzed under two cases: instantaneous external force and asymmetric continuous load. The simulation results are shown in Fig. 12 and Fig. 13, respectively. The system responses are evaluated through two parameters of vehicle tilt angle and reaction wheel velocity.

Under the instantaneous external force, it is observed that from Fig. 12:

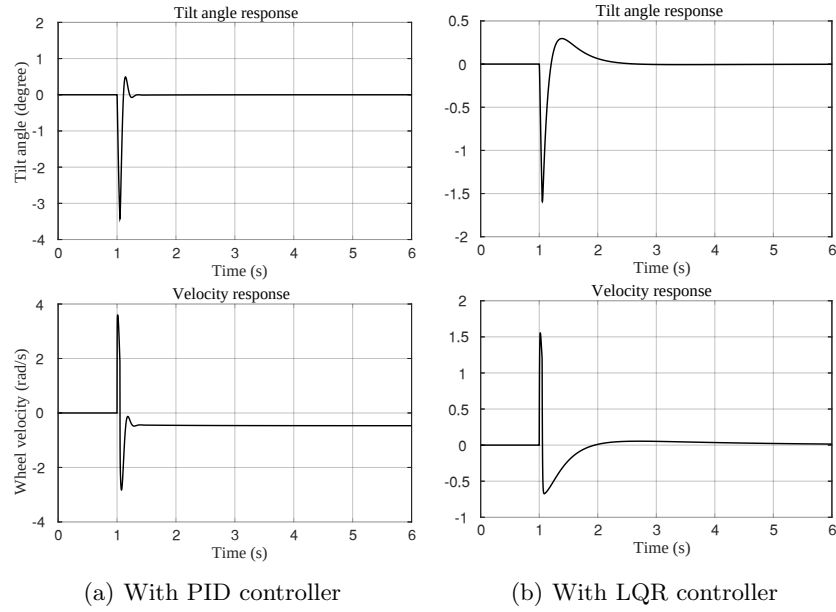


Fig. 12. System responses under instantaneous external force

- Both PID and LQR have restored the system to the initial balancing position.
- The tilt angle using PID responses faster than that of LQR.
- The wheel velocity using LQR gradually returns to 0 rad/s, while the wheel velocity using PID converges to another value that is different from the initial velocity of 0 rad/s.

It is observed that from Fig. 13:

- The PID controller is not able to pull the model back to the balancing position. In detail, the tilt angle retains its position at 3.2° before gradually falling to the side carrying the asymmetric continuous load. This can be explained as time goes by, the speed response decreases continuously in order to retain the position of the model and then reaches its maximum speed where saturation happens. As a result, the motor is no longer able to generate torque.
- The LQR controller is able to bring the model back to the balancing position. Specifically, after falling to the side carrying the load, the angle response is brought up to the 0° position due to the torque generated by the change in speed of the motor. Sequentially, the speed response gradually increases, preventing the motor from reaching its saturation state and keeping self-balancing around its equilibrium position.

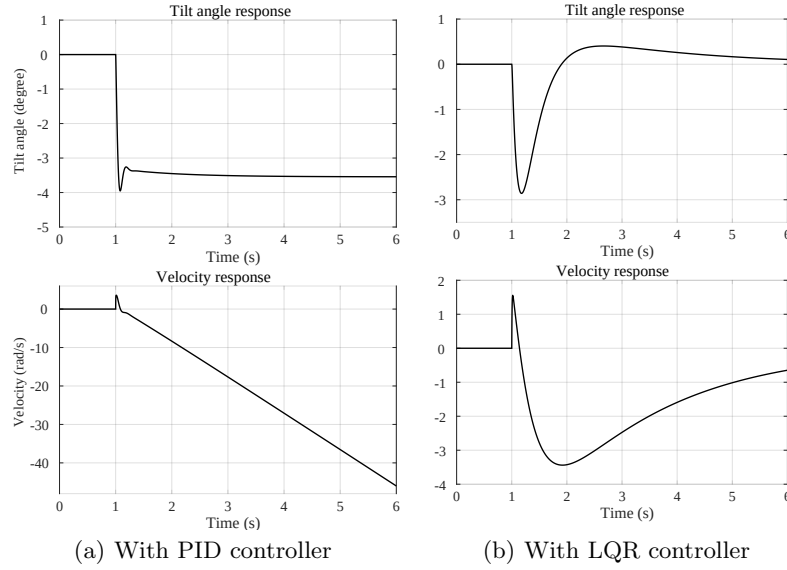


Fig. 13. System responses under asymmetric continuous force

5 Experimental Results on The Prototype

5.1 Fine-tune LQR coefficients

According to the above section, the state weighting parameters can be achieved as follows:

$$Q_{11} = 4000, \quad Q_{22} = 150, \quad Q_{33} = 1, \quad Q_{44} = 5$$

Sequentially, the feedback gain matrix K is determined from MATLAB with the given state weighting parameters above:

$$K = [-203.3016 \quad -19.5979 \quad -1.0000 \quad -2.8459]$$

After that, the calculated feedback gain matrix is substituted into the program of the physical prototype to test its performance. However, the prototype operated in the real environment faces a lot of noise and disturbance coming from various sources. Therefore, each index in the gain matrix will be carefully tuned to optimize their response. As a result, the final feedback gain matrix is obtained with some adjustments as indicated below:

$$K = [-150 \quad -20 \quad -0.1 \quad -2]$$

5.2 Evaluate Results From Various Conditions

To verify the LQR controller performance in various operating scenarios, a complete prototype shown in Fig. 14 is created by assembling all the components in Section 3 and Section 4.

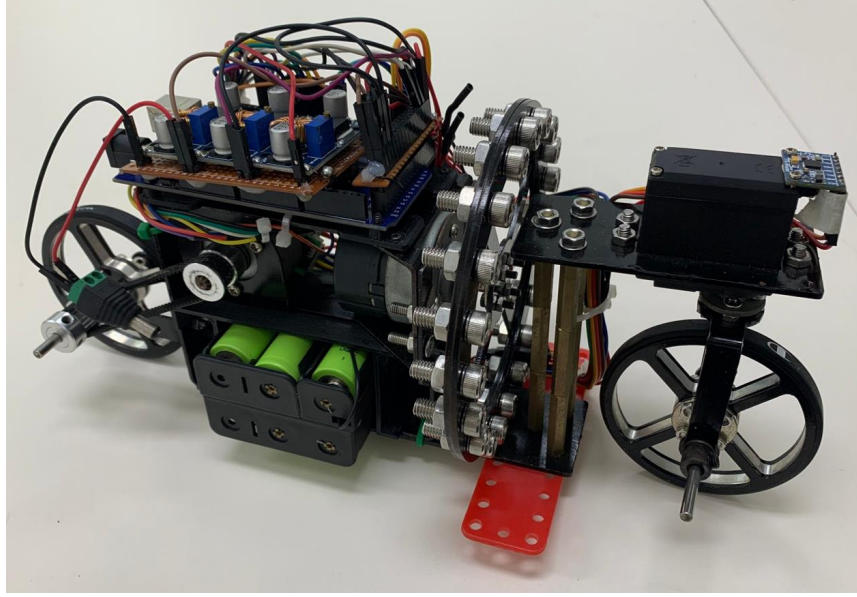


Fig. 14. Physical prototype balancing performance

The balancing performance is verified through two plots: the tilt angle of the vehicle and the wheel velocity during its balancing operation. In the following experiments, the wheel velocity values are sampled at 100 Hz and their average velocity is computed for every 20 samples for data interpretation.

- Operating scenario 1: Without external force
 - Condition: The static system is held in balance initially. There is no disturbance during its self-balance.
 - Observation: The experimental result is shown in Fig. 15, in which it is clearly observed that the wheel velocity always oscillates about a small approximate value of -3 rad/s and the vehicle is vertically maintained at about 0.5° .
- Operating scenario 2: With instantaneous transverse force
 - Condition: The static vehicle is also hold in balance initially. Then, an instantaneous external force is applied to the system sideways at the time of 0.75 second as seen in Fig. 16.
 - Observation: At this point of time, the vehicle is pushed to one side and in a short time later, the wheel velocity is changed drastically to restore its balance from the time 1.5 seconds onward.
- Operating scenario 3: With asymmetric continuous load
 - Condition: The vehicle self-balances without disturbance around 1.5° during the first 0.75 second. After that, an asymmetric continuous load is placed on one side of system.

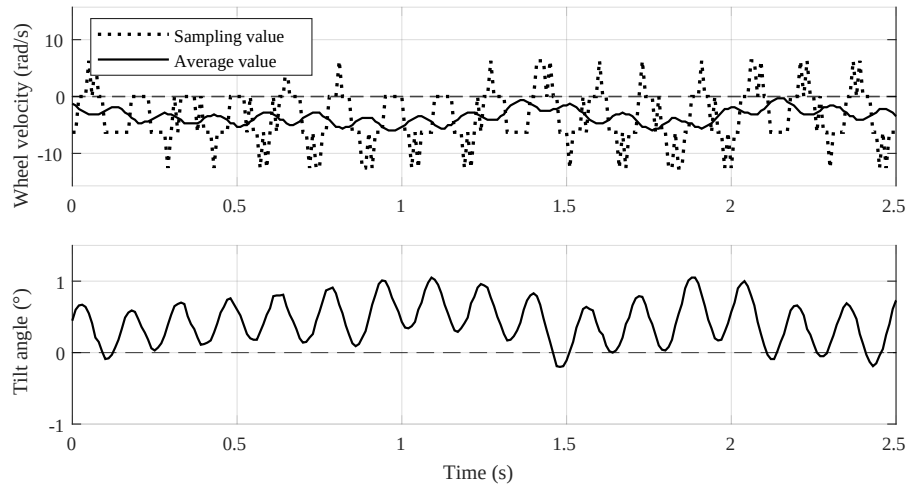


Fig. 15. Balancing performance without external force

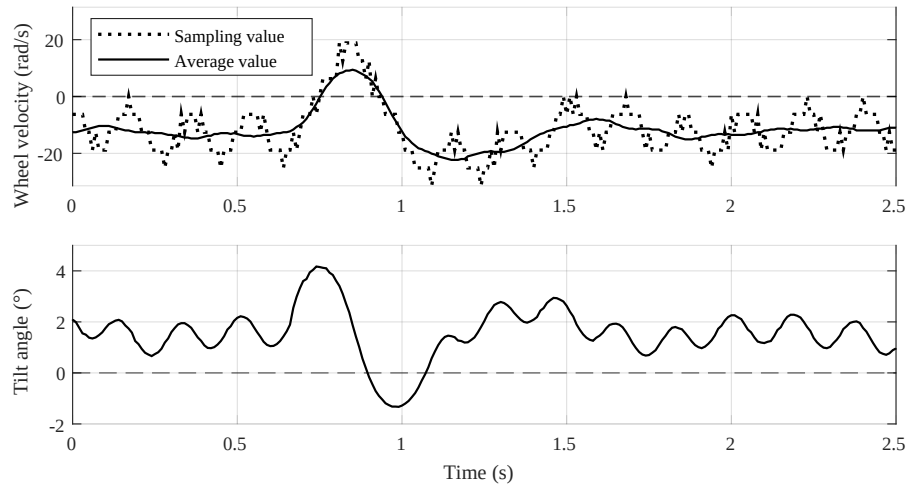


Fig. 16. Balancing performance with instantaneous transverse force

- Observation: The load causes the vehicle to fall to -0.8° as shown in Fig. 17. As a result, the average wheel velocity rapidly decreases to almost -50 rad/s, which generates enough torque to pull the system back to its balancing position about 3° .

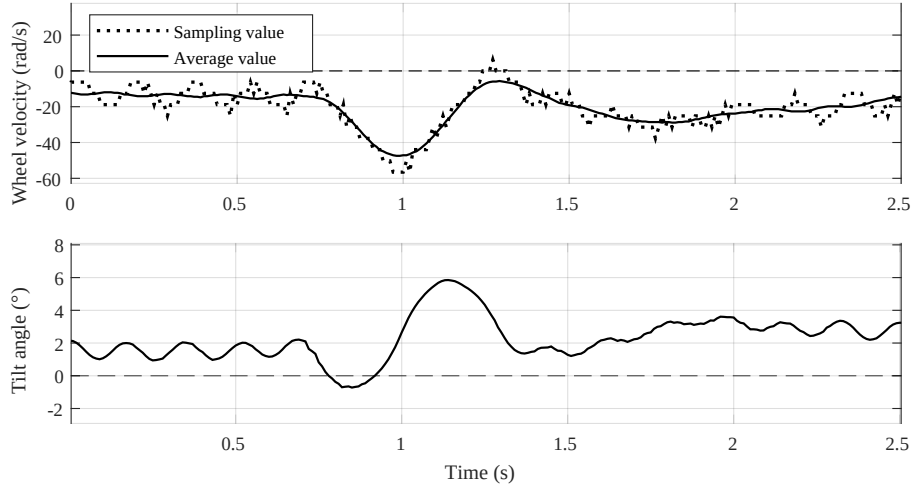


Fig. 17. Balancing performance with asymmetric continuous load

- Operating scenario 4: Moving forward
 - Condition: The vehicle is moving in a straight line on table surfaces with the translational velocity of 0.3 m/s. As the table edges are noncoplanar, the vehicle has to pass the rough path without falling.
 - Observation: The experimental result is demonstrated in Fig. 18 in which the vehicle is swung about 2.5° . In addition, the average wheel velocity is always around -20 rad/s.
- Operating scenario 5: In circular motion
 - Condition: For the first 0.5 second, the vehicle stationarily maintains its balance around 2° without external force. Sequentially, it speeds up to 0.3 m/s with the steering angle of 30° to move in circular motion.
 - Observation: The rapid change in translational velocity causes the tilt angle to fluctuate between -1.8° and 4.5° as indicated in Fig. 19. After that, the system obtains its balance again with the tilt angle gradually falls to 2° . It is noticed that the wheel velocity is maintained around 20 rad/s.

With all the experimental results as discussed above, the LQR controller can be used effectively for the self-balancing vehicle as the tilt angle of the vehicle and the wheel velocity always converges to small values.

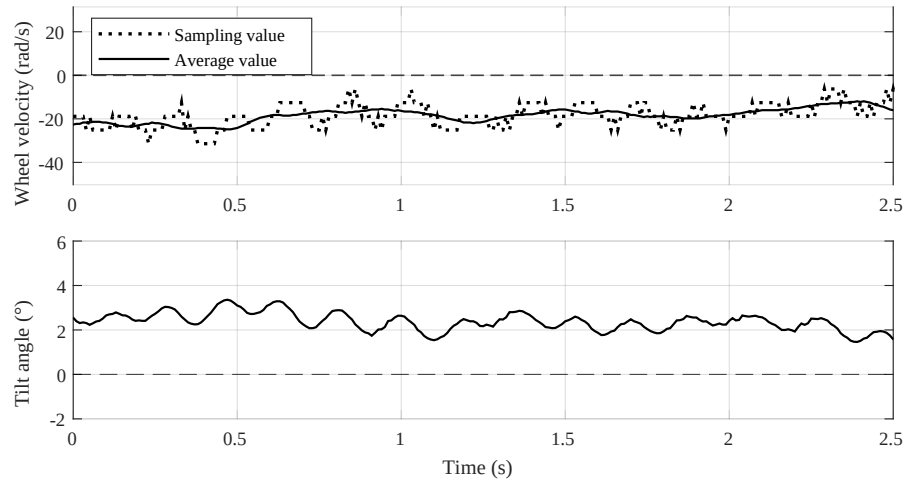


Fig. 18. Balancing performance when moving forward

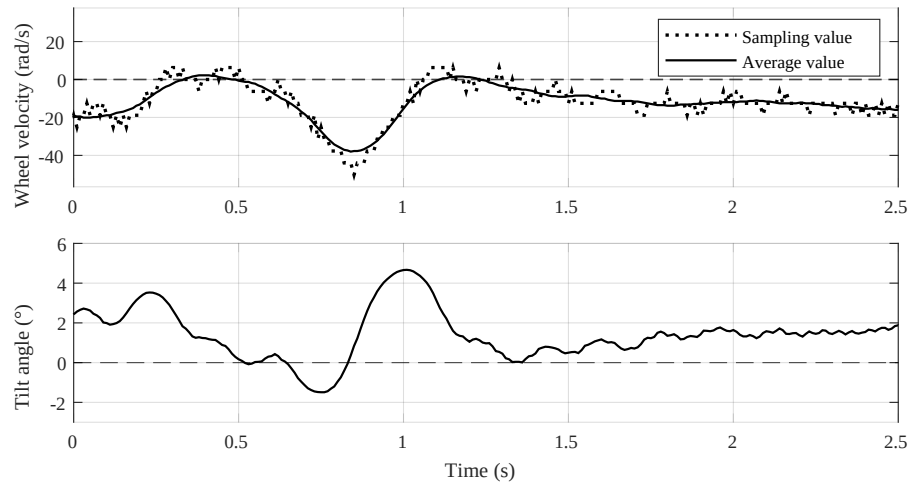


Fig. 19. Balancing performance in circular motion

6 Conclusions

This paper has highlighted the mathematical models of the motor and the vehicle. Study on the effects of some key parameters onto the model is effectively utilized to properly design the reaction wheel. Moreover, IMU sensor performance with Kalman filter has been successfully implemented. PID and LQR control algorithms have been explored through simulation. Consequently, the result has indicated that the LQR controller provides more stable responses. In addition, LQR algorithm has been validated on the physical prototype in various scenarios where the tilt angle is often within $\pm 3^\circ$. Further development will emphasize on implementing the vision-based system on the self-balancing motorcycle to autonomously track line paths.

References

1. J.Yi et al., Trajectory tracking and balance stabilization control of autonomous motorcycle, International Conference on Robotics and Automation, Orlando 2006.
2. P. Seekhao, K. Tungpimolrut, and M. Parnichkun, Development and control of a bicycle robot based on steering and pendulum balancing, Journal of Mechatronics, Vol. 69, 2020.
3. A. Mohan et al., Self Balancing of Two Wheeler, International Journal of Innovative Science and Research Technology, Vol. 3(3), 2018.
4. R. Hartono et al., Design of Self-balancing Model-size Electrical Motorbike Robot Using Control Moment Gyroscope, Journal of Sensors and Materials, Vol. 35, 2023.
5. Yadhukrishnan and I.R. Saxena, Self-balancing of a Two wheeler using Gyroscopic effect, International Journal of Advance Research, Ideas and Innovations in Technology, Vol. 6(5), 2020.
6. R.R. George et al., Study on Self Balancing Motorcycle Using the Concept of Reinforcement Learning, International Research Journal of Engineering and Technology, Vol. 7(4), 2020.
7. D. Vasilevski, D. Pecioski, and S.D. Markovska, Design and Implementation of Self-Balancing Motorcycle, Journal of Mechanical Engineering, Vol. 41(2), 2023.
8. Q. K. Ho and C. B. Pham, Study on Inertia Wheel Pendulum Applied to Self-Balancing Electric Motorcycle, The 4th International Conference on Green Technology and Sustainable Development, Viet Nam, 2018.